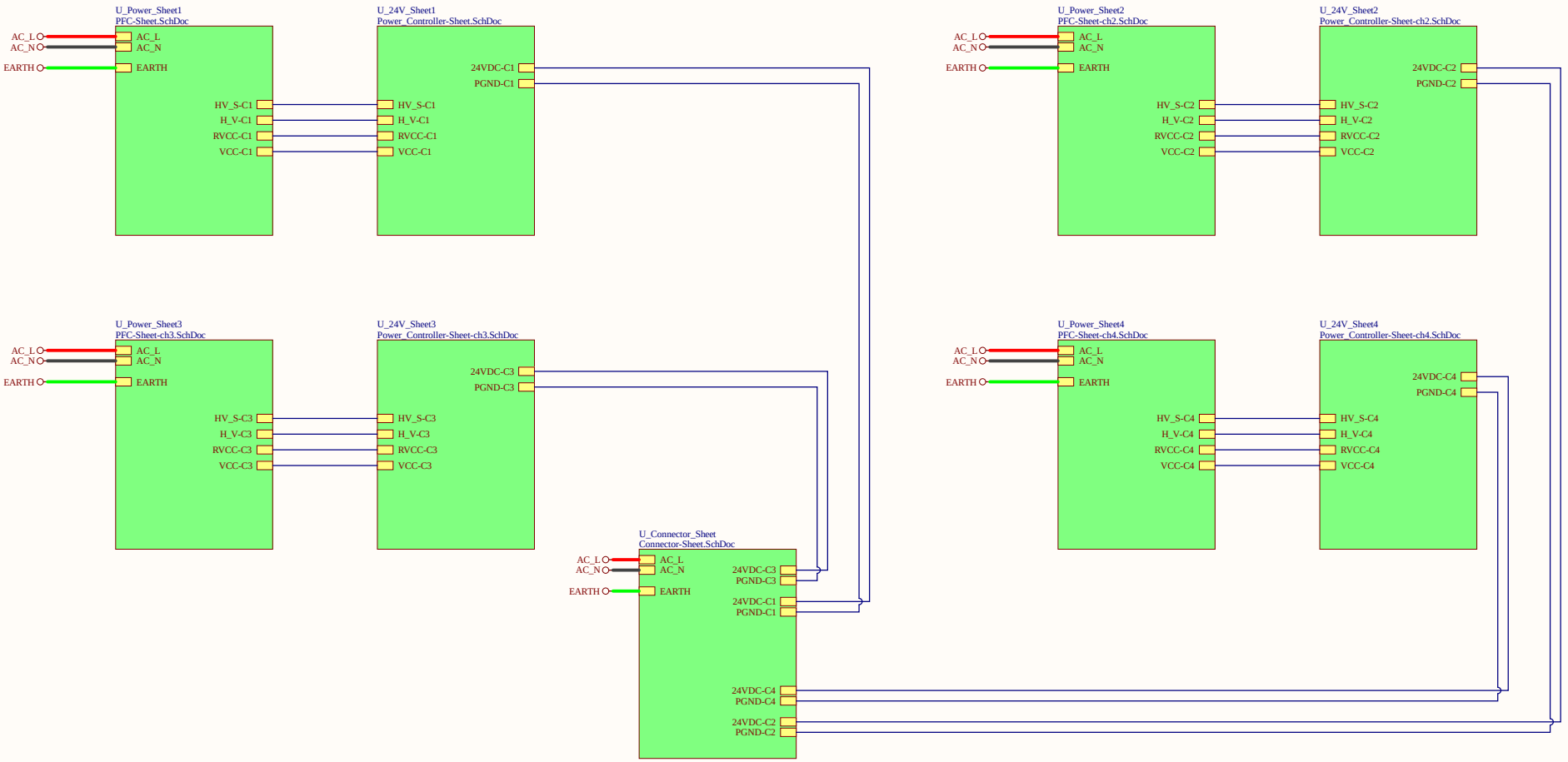
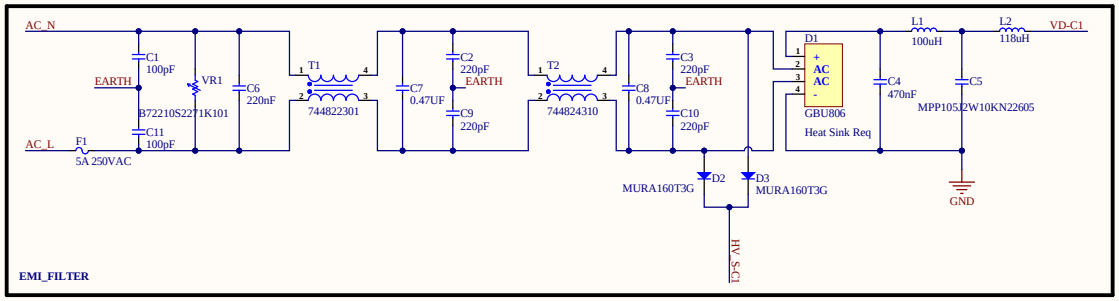
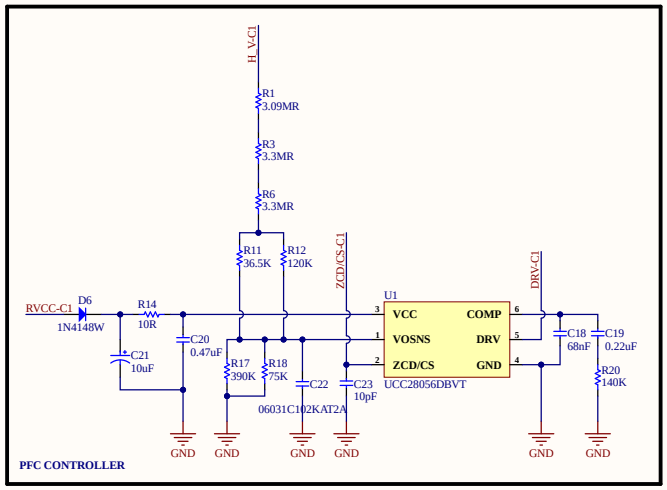


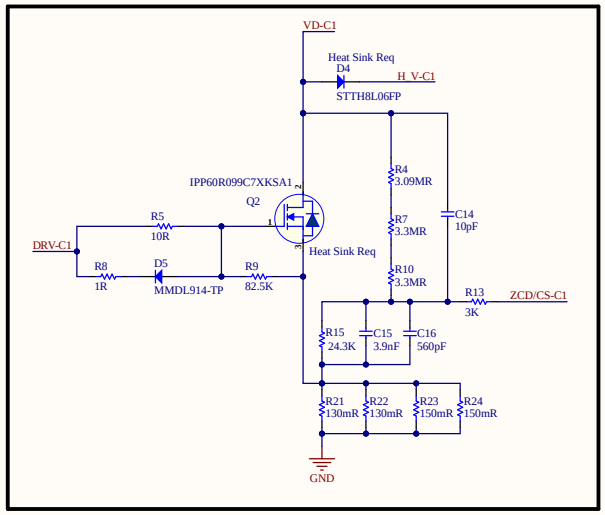
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File:	24V-10A-Power-Board-v2.0.PrjPcb
Date: 2/12/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10

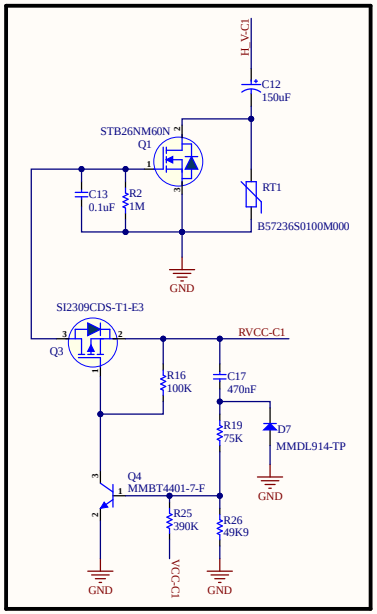
Note: Use insulation / thermal pad between transistors/diode and heatsink

Note:
A 75mm x 30mm x 2.54mm heatsink is applied to D4 , D1 & Q2









HV_S-C1 — HV_S-C1
H_V-C1 — H_V-C1
RVCC-C1 — RVCC-C1
VCC-C1 — VCC-C1

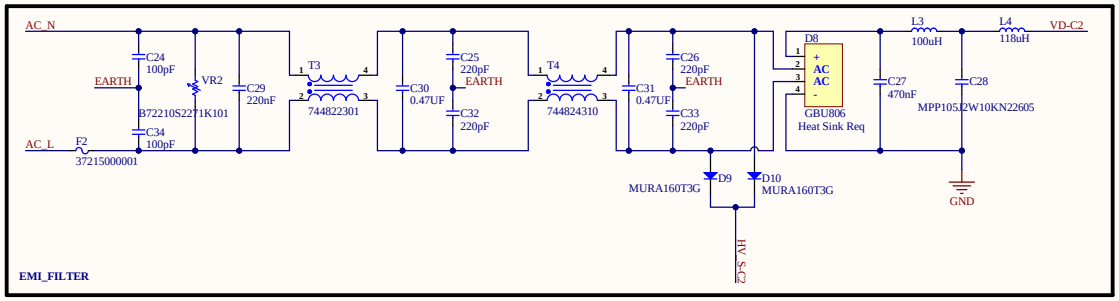
AC_L — AC_L
EARTH — EARTH
AC_N — AC_N

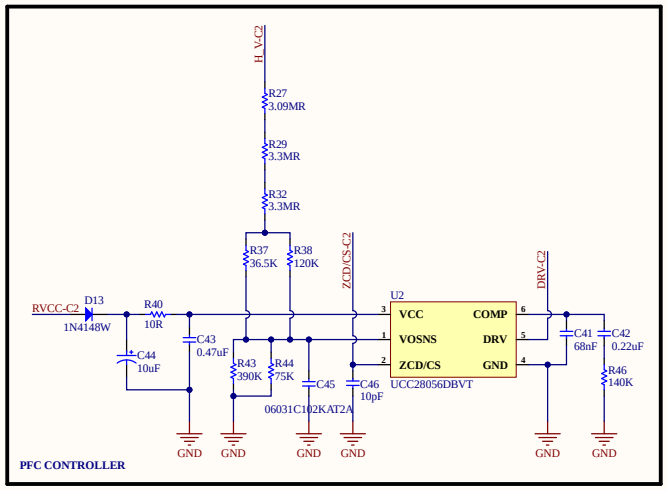
eSpark Consulting Group

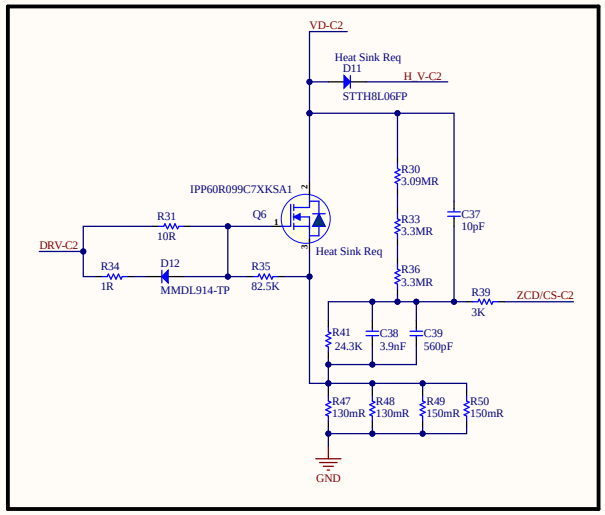
Title: PFC-Sheet.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 6/19/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 10

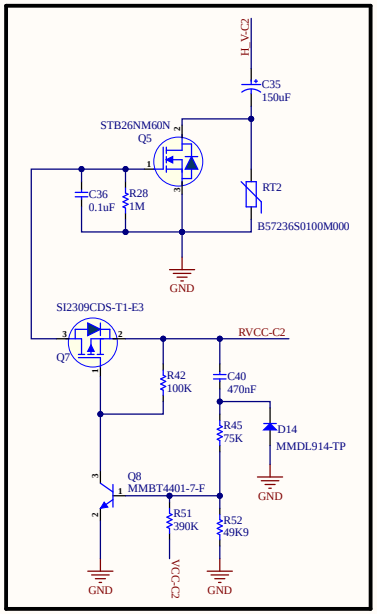
Note: Use insulation / thermal pad between transistors/ diode and heatsink

Note:
A 75mm x 30mm x 2.54mm heatsink is applied to D11 , D8 & Q6









HV_S-C2 — HV_S-C2
H_V-C2 — H_V-C2
RVCC-C2 — RVCC-C2
VCC-C2 — VCC-C2

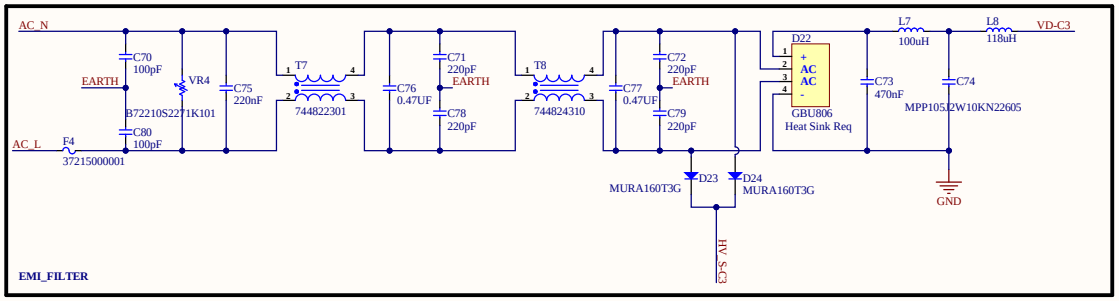
AC_L — AC_L
EARTH — EARTH
AC_N — AC_N

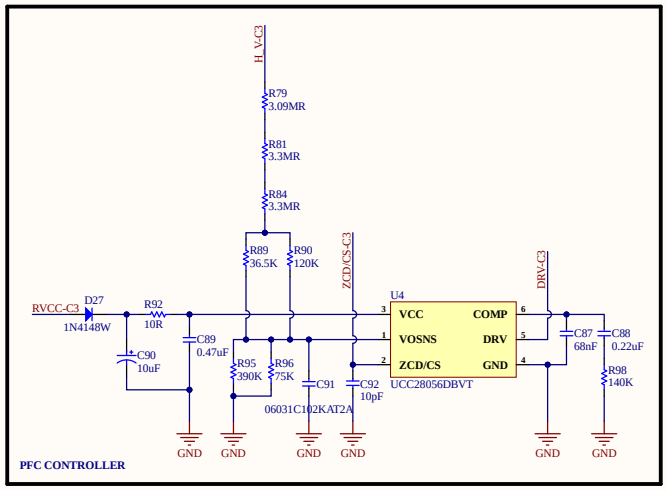
eSpark Consulting Group

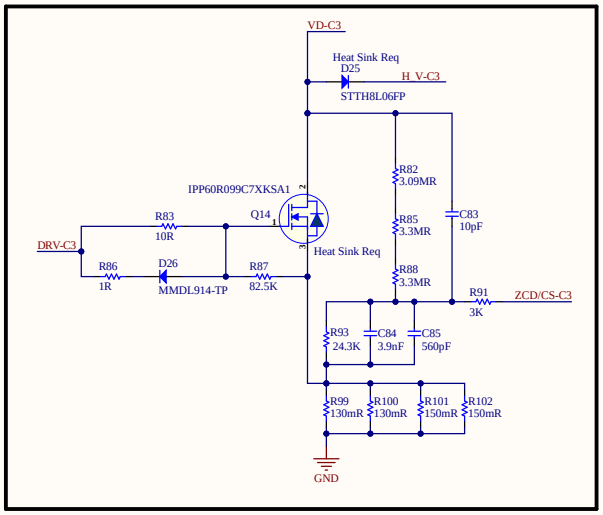
Title: PFC-Sheet-ch2.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 6/19/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 10

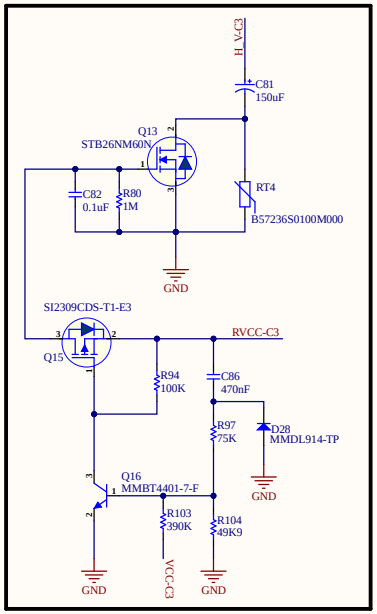
Note: Use insulation / thermal pad between transistors/ diode and heatsink

Note:
A 75mm x 30mm x 2.54mm heatsink is applied to D22 , D25 & Q14









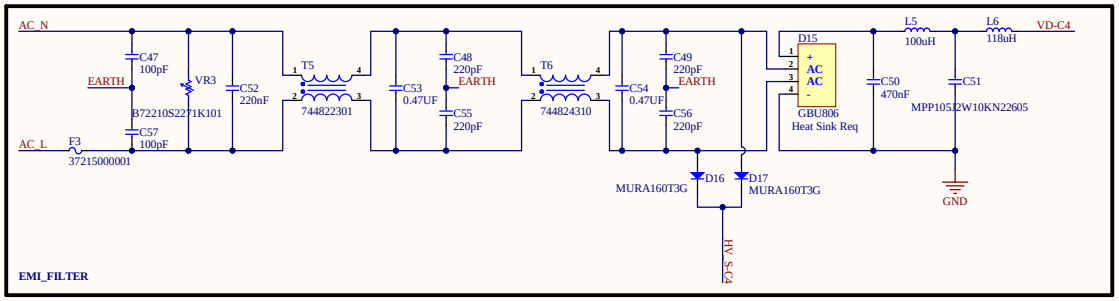
HV_S-C3 — HV_S-C3
H_V-C3 — H_V-C3
RVCC-C3 — RVCC-C3
VCC-C3 — VCC-C3

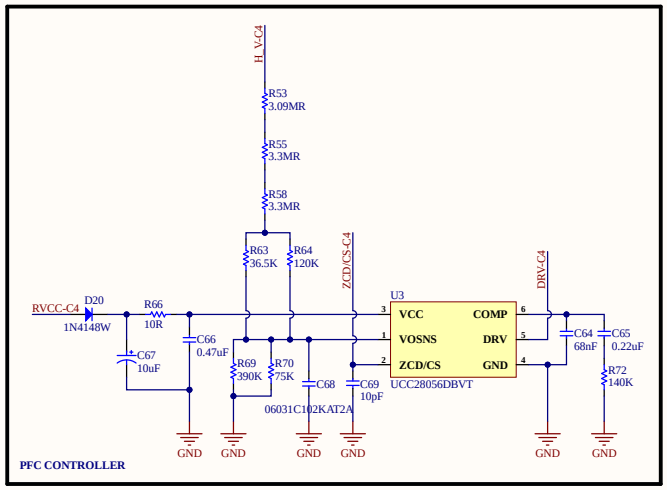
AC_L — AC_L
EARTH — EARTH
AC_N — AC_N

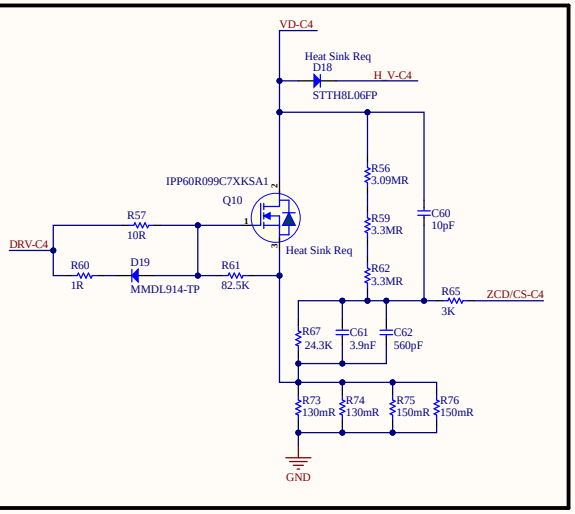
Title: PFC-Sheet-ch3.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 6/19/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 10

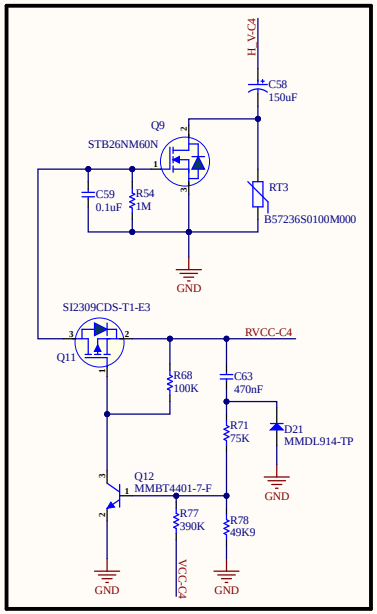
Note: Use insulation / thermal pad between transistors/ diode and heatsink

Note:
A 75mm x 30mm x 2.54mm heatsink is applied to D15 , D18 & Q10







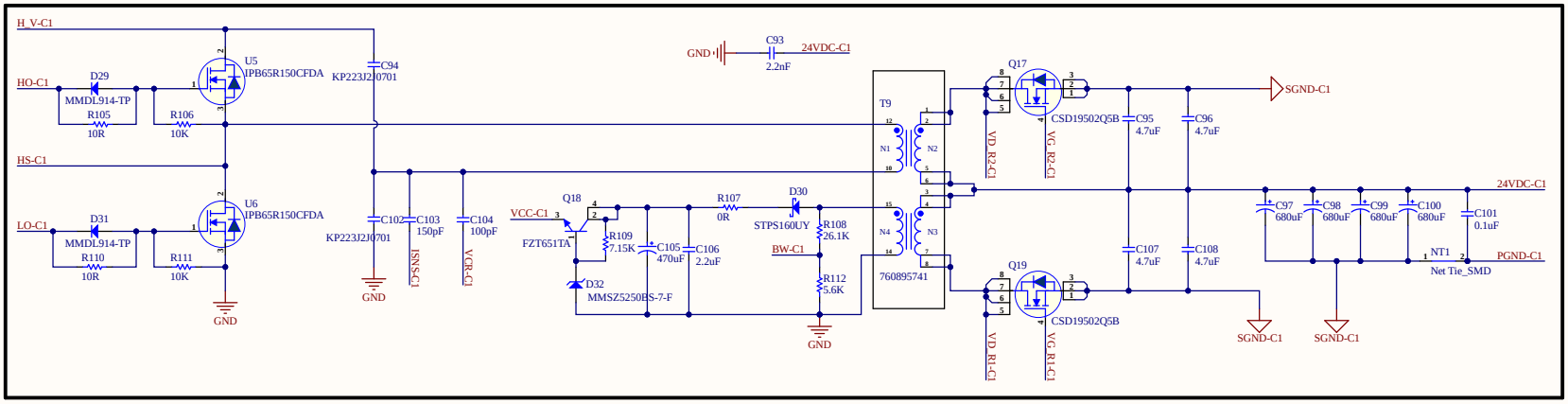


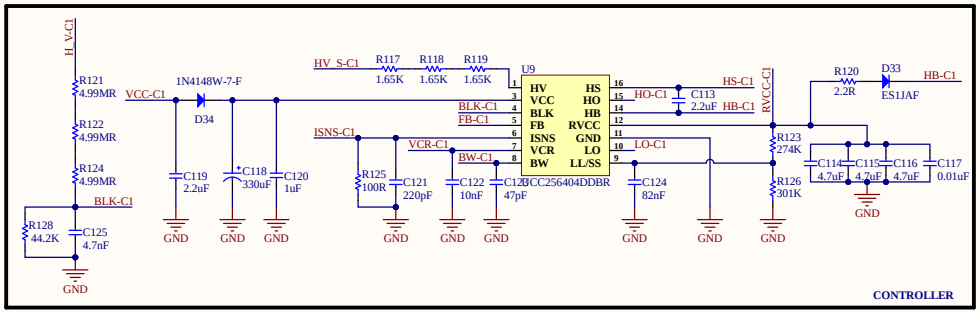
HV_S-C4 — HV_S-C4
H_V-C4 — H_V-C4
RVCC-C4 — RVCC-C4
VCC-C4 — VCC-C4

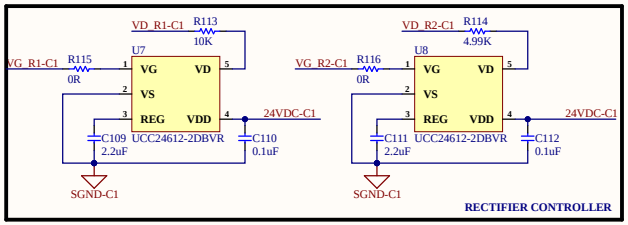
AC_L — AC_L
EARTH — EARTH
AC_N — AC_N

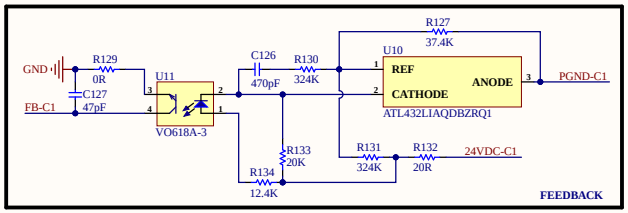
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Title: PFC-Sheet-ch4.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 6/19/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 10









24VDC-C1 — 24VDC-C1

PGND-C1 — PGND-C1

HV_S-C1 — HV_S-C1

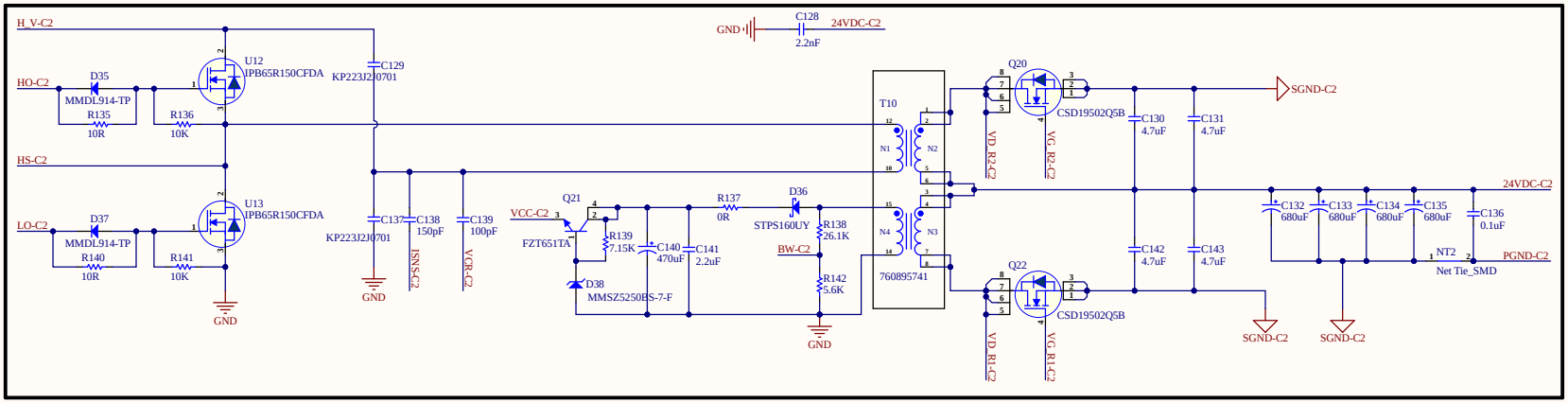
H_V-C1 — H_V-C1

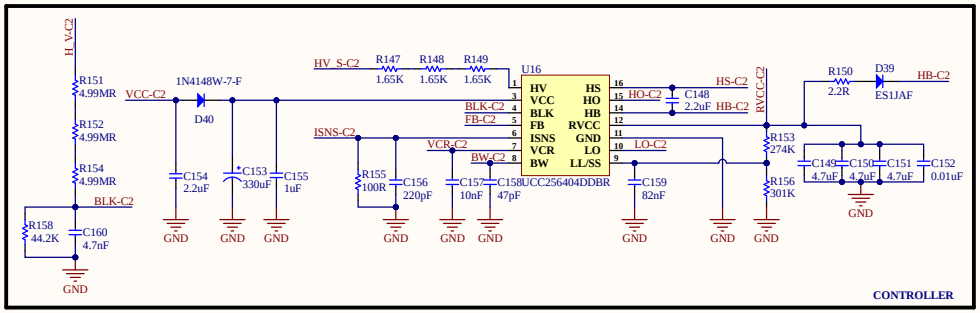
RVCC-C1 — RVCC-C1

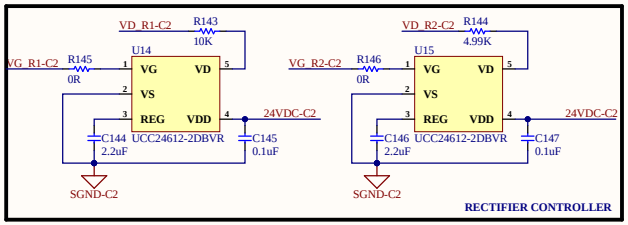
VCC-C1 — VCC-C1

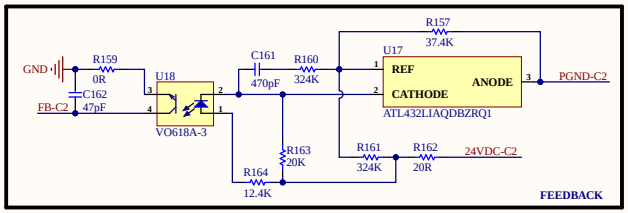
eSpark Consulting Group

Title: Power_Controller-Sheet.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 3/21/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10









24VDC-C2 — 24VDC-C2

PGND-C2 — PGND-C2

HV_S-C2 — HV_S-C2

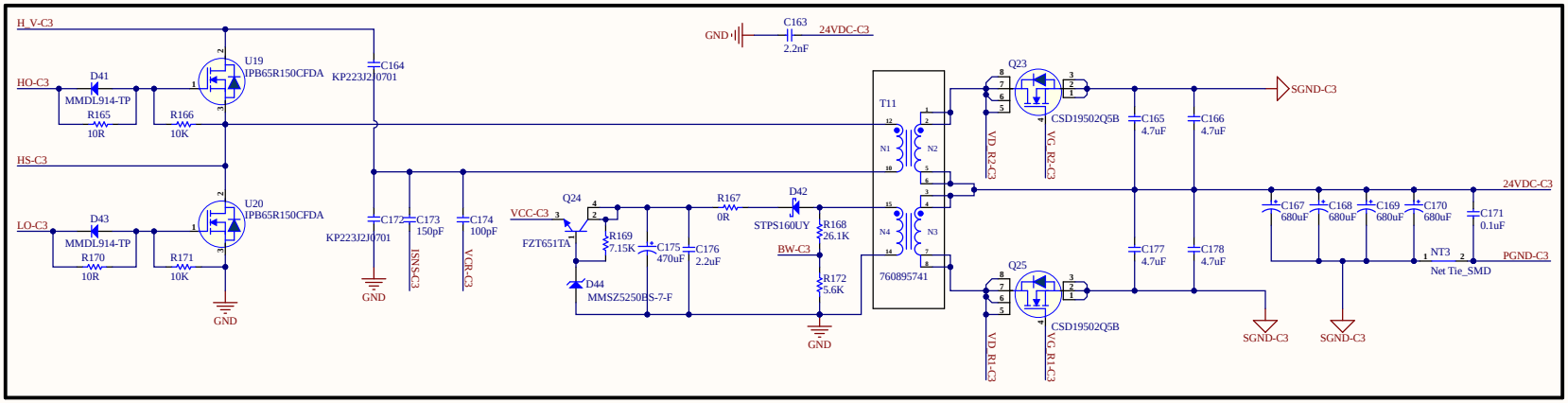
H_V-C2 — H_V-C2

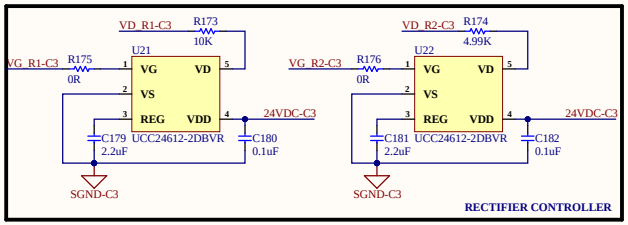
RVCC-C2 — RVCC-C2

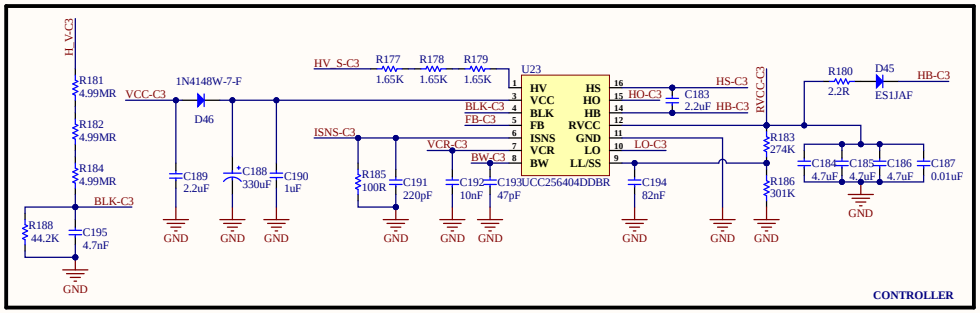
VCC-C2 — VCC-C2

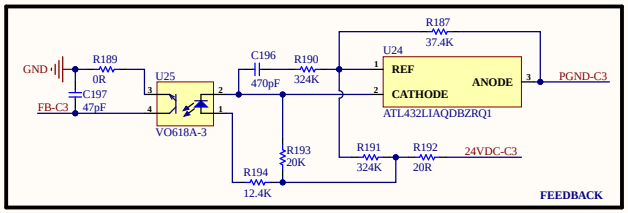
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Title: Power_Controller-Sheet-ch2.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 3/21/2025	Page Num: 7
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24VDC-C3 — 24VDC-C3

PGND-C3 — PGND-C3

HV_S-C3 — HV_S-C3

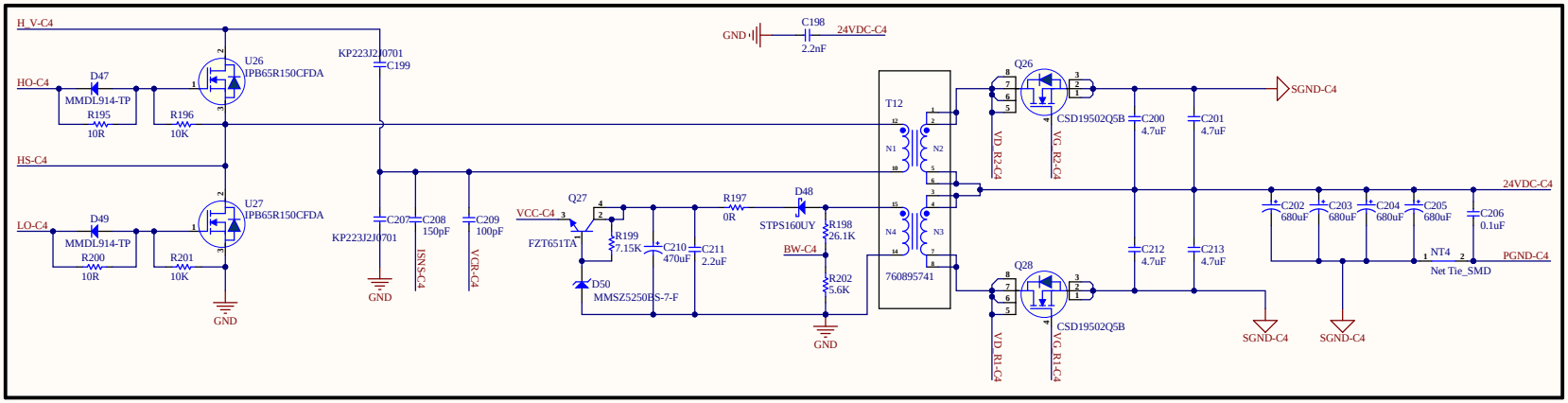
H_V-C3 — H_V-C3

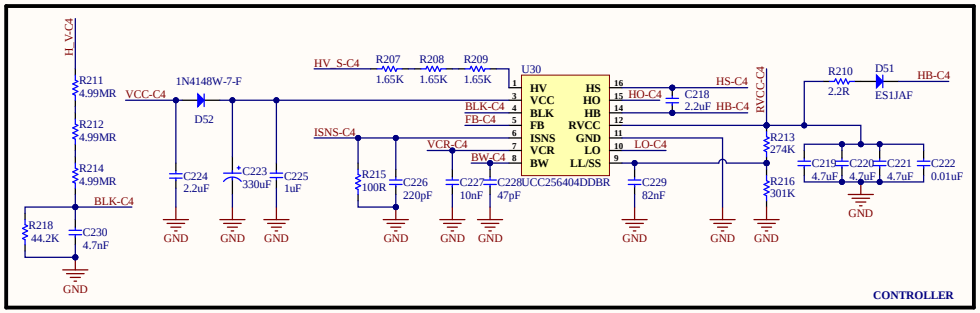
RVCC-C3 — RVCC-C3

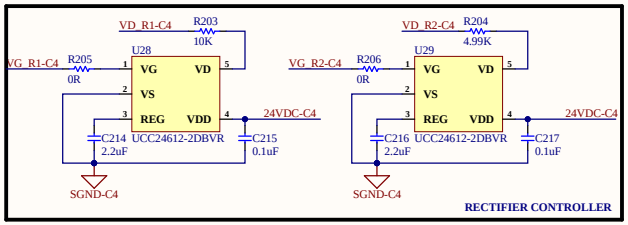
VCC-C3 — VCC-C3

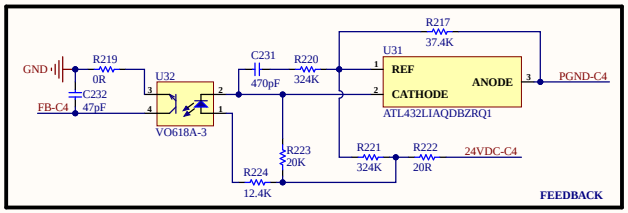
eSpark Consulting Group

Title: Power_Controller-Sheet-ch3.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 3/21/2025	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 10









24VDC-C4 — 24VDC-C4

PGND-C4 — PGND-C4

HV_S-C4 — HV_S-C4

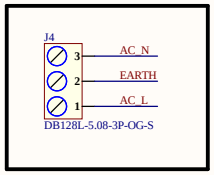
H_V-C4 — H_V-C4

RVCC-C4 — RVCC-C4

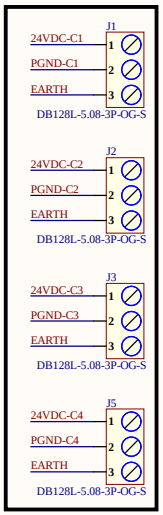
VCC-C4 — VCC-C4

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Title: Power_Controller-Sheet-ch4.SchDoc	
File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 3/21/2025	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 10



AC 110 ~ 230 VAC In



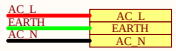
24VDC 10A Out











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File: 24V-10A-Power-Board-v2.0.PrjPcb	
Date: 3/21/2025	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 10